



JOHNSON CONTROLS, INC.
6935 Appling Farms Parkway
Memphis, TN 38133
(901)383-5757

Report of Eddy Current Inspection

Manufacturer: Trane

Model: CVHE089FA

Serial: L92A00803

Location: HERSHEY FOODS
975 KANSAS STREET
SOUTH CHILLER ROOM
MEMPHIS, TN 38106

Inspected: December 16, 2013

Inspected By: MARTIN W. GOINS, LEVEL III

Reviewed By:

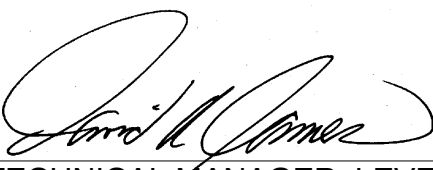

TECHNICAL MANAGER, LEVEL III

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Vessel Information

Manufacturer	Model	Style	Serial Number	Type
Trane	CVHE089FA	Hermetic	L92A00803	Centrifugal

Condenser	
TestEnd	Right Hand Facing Controls
Tube Count	442
Tube Type	Skip Fin IE
Tube Material	Copper
OD	1.000
*NWT/Under Fins	.028
*NWT/Bell/Land	.053
#/Type Support	3 Mild Steel
Tube Numbering	Left to Right
Row Numbering	Top to Bottom
Tube Length +/- 2	180 Inches

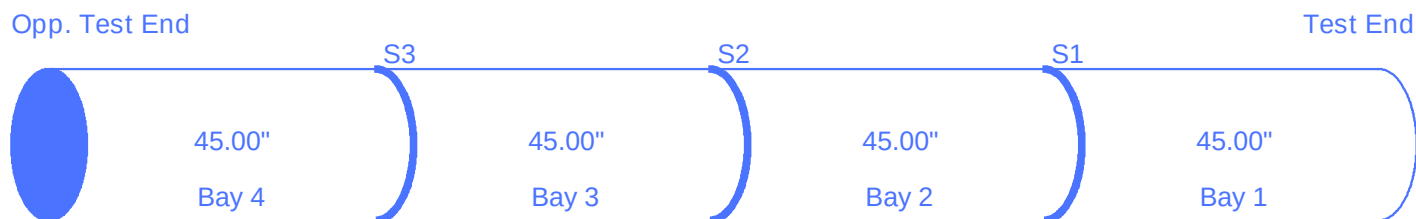
Evaporator	
TestEnd	Right Hand Facing Controls
Tube Count	241
Tube Type	Skip Fin IE
Tube Material	Copper
OD	1.000
*NWT/Under Fins	.028
*NWT/Bell/Land	.053
#/Type Support	4 Mild Steel
Tube Numbering	Left to Right
Row Numbering	Top to Bottom
Tube Length +/- 2	180 Inches

Analyst: MARTIN W. GOINS, LEVEL III

* Nominal Wall Thickness

Vessel Bay Length Information

Condenser (Length = 180 inches)
S = Intermediate Support



Bay 4	45.00"
Bay 3	45.00"
Bay 2	45.00"
Bay 1	45.00"

Evaporator (Length = 180 inches)
S = Intermediate Support



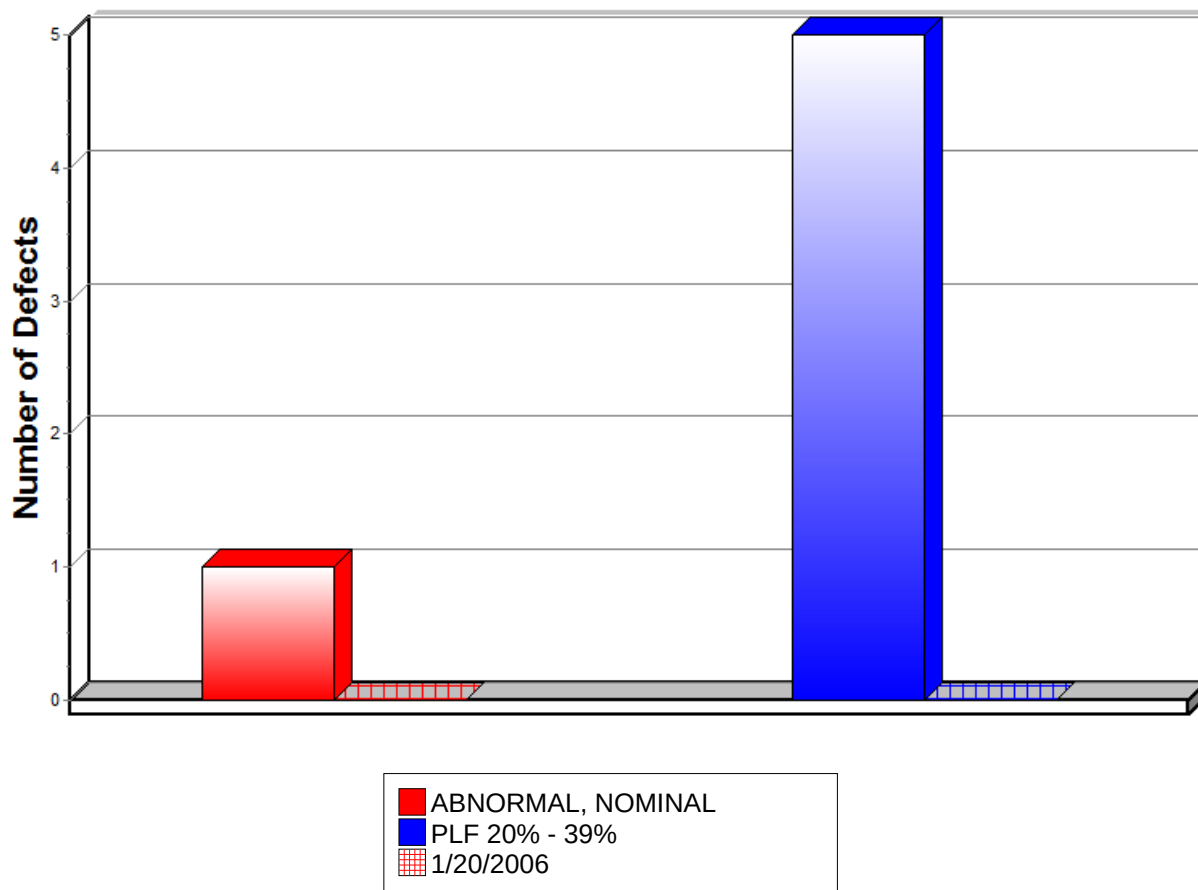
Bay 5	36.00"
Bay 4	36.00"
Bay 3	36.00"
Bay 2	36.00"
Bay 1	36.00"

Defect Summary/Comparison

Comparison of Tests Performed

12/16/2013 1/20/2006

Condenser Defects



Location	Model	Serial Number
HERSHEY FOODS	CVHE089FA	L92A00803

Note: The Graph will indicate a Comparison Analysis when the unit has been previously tested by TAI Services.

Summary of Inspection

An eddy current tube inspection was performed as part of a preventive maintenance program with the following results.

Condenser: 442 Tubes		
Significant/Measurable Indications	Number of Tubes Marked	Percent of Bundle
PLF 20% - 39%	5	1.13
ABNORMAL, NOMINAL	1	.23
Totals	6	1.36

Evaporator: 241 Tubes		
Significant/Measurable Indications	Number of Tubes Marked	Percent of Bundle
NO MEASURABLE DEFECTS		
Totals	0	.00

Recommendations

An eddy current inspection was performed on the tubes in this machine. This test was performed using accepted eddy current test methods for the inspection of in-service tubing. It should be noted that Eddy Current is not a leak detection method. The possibility does exist that tubes could contain defects and/or leaks which are not detectable. If leaks are suspected, we recommend a pressure test be used to identify the leaking tubes.

The following suggested repair actions are based on accepted industry standards. After removing sample tubes to confirm the inspection results, a determination of corrective action should be made by the repair agency and end user. Only these parties have knowledge of the critical applications and long-term use of the equipment. If plugging is selected over replacement, both efficiency and capacity should be considered.

CONDENSER:

Tubes indicating Possible Longitudinal Flaws (PLF) require no corrective action at the present time. However, these defects can be progressive and should be monitored for defect growth.

The tubes indicating Abnormal Nominal Indications require no corrective action at this time. The indications do not appear to be that of significant tube damage. However, the exact cause of the indication is unknown.

EVAPORATOR:

There were no measurable defects noted during this inspection.

RE-INSPECTION RECOMMENDATIONS:

We recommend that a follow-up inspection be performed on these vessels as follows:

Condenser: 16 December 2016

Evaporator: 16 December 2016

A copy of this report should be retained in your files to be used for comparison at that time.

If you should have any questions concerning this report, or if we may be of further assistance, please feel free to call upon us.

Data Sheet

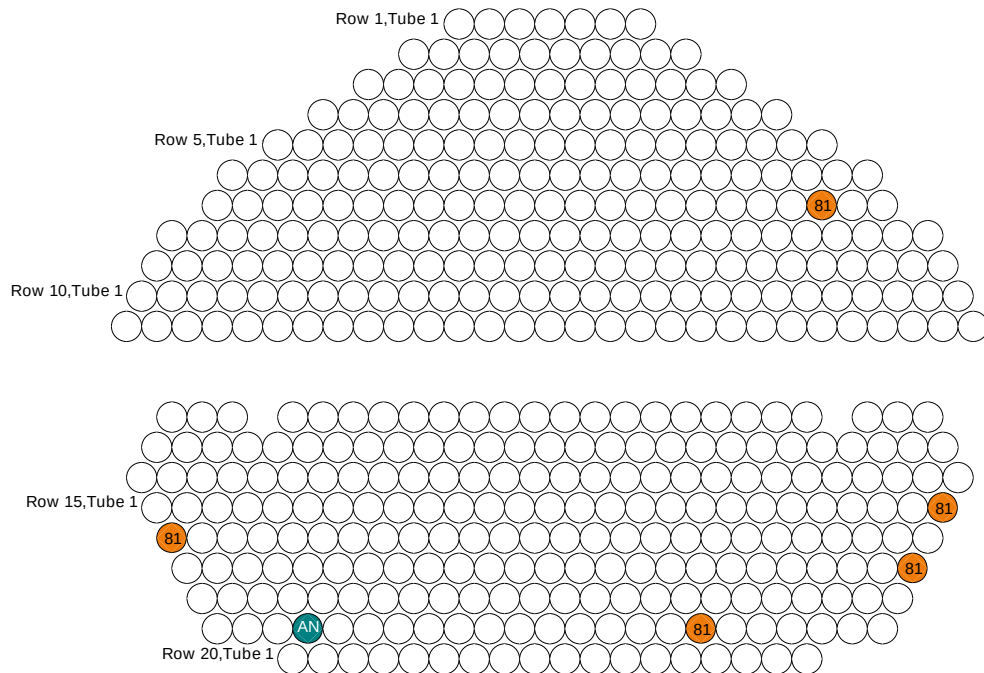
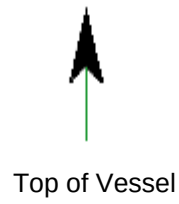
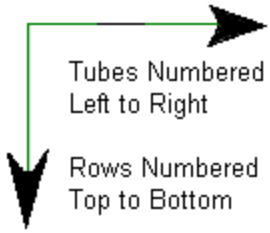
Location	Model	Serial Number	Date
HERSHEY FOODS	CVHE089FA	L92A00803	December 16, 2013
975 KANSAS STREET			
MEMPHIS, TN 38106			

Row	Tube	Description	Area	Action Req.
SET UP CALIBRATE & STARTED				
CONDENSER 12/16/2013 07:02 am				
7	21	PLF 20% - 39%	B03	
15	27	PLF 20% - 39%	B04	
17	25	PLF 20% - 39%	B04	
19	17	PLF 20% - 39%	B04	
16	1	PLF 20% - 39%	B04	
19	4	ABNORMAL, NOMINAL	OTE	
CALIBRATION CHECK & COMPLETED				
CONDENSER 12/16/2013 09:04 am				
SET UP CALIBRATE & STARTED				
EVAPORATOR 12/16/2013 09:05 am				
NO MEASURABLE DEFECTS				
CALIBRATION CHECK & COMPLETED				
EVAPORATOR 12/16/2013 10:27 am				

Condenser Section

S/N L92A00803

Right Hand Facing Controls



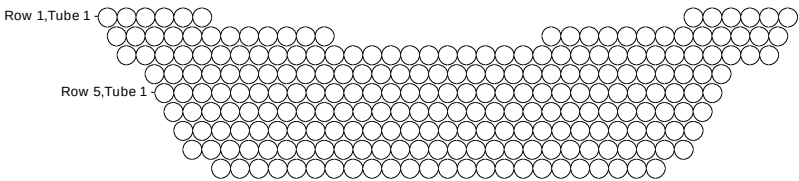
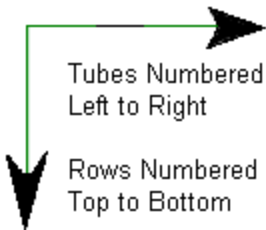
81 = PLF 20% - 39%

AN = ABNORMAL, NOMINAL

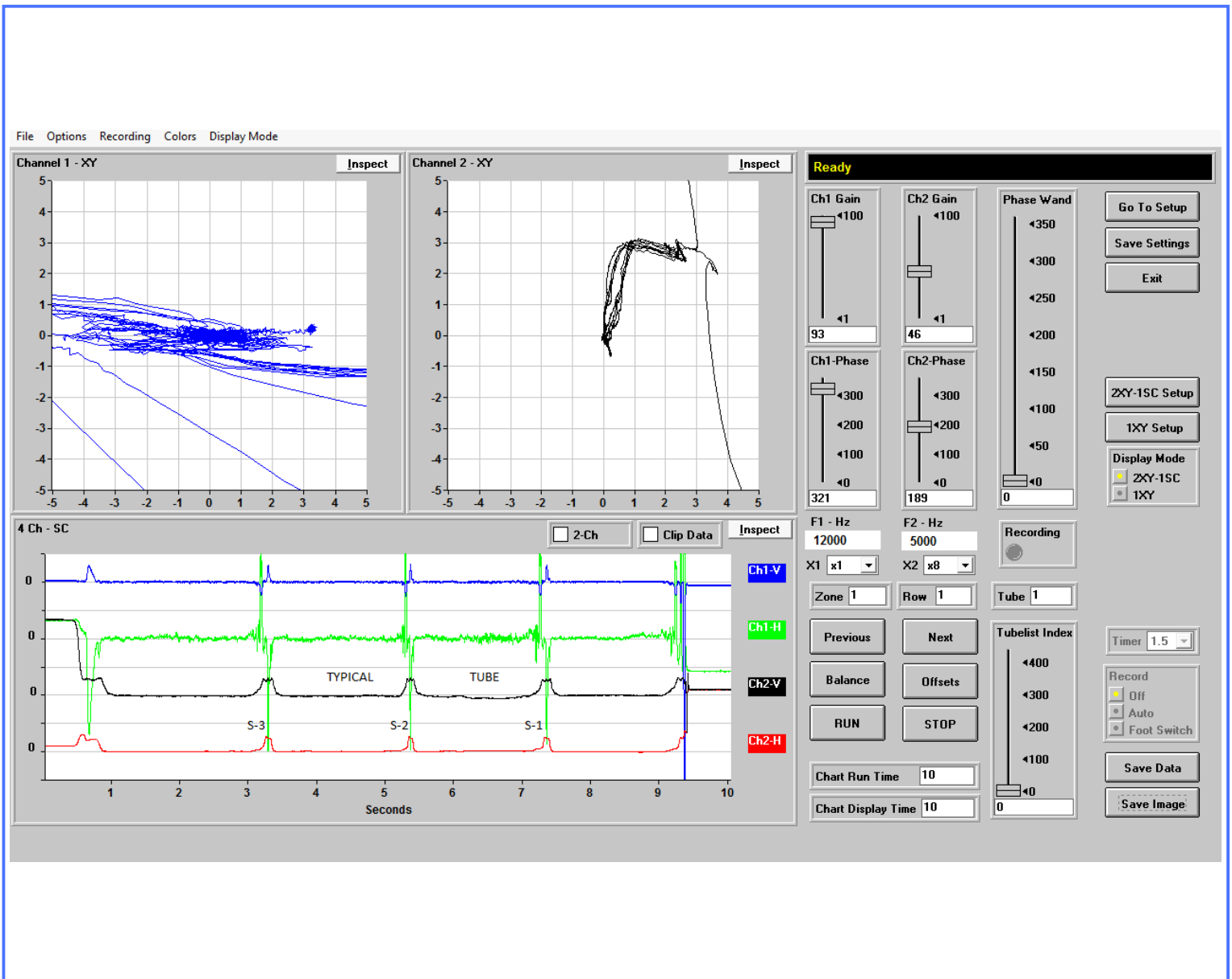
Evaporator Section

S/N L92A00803

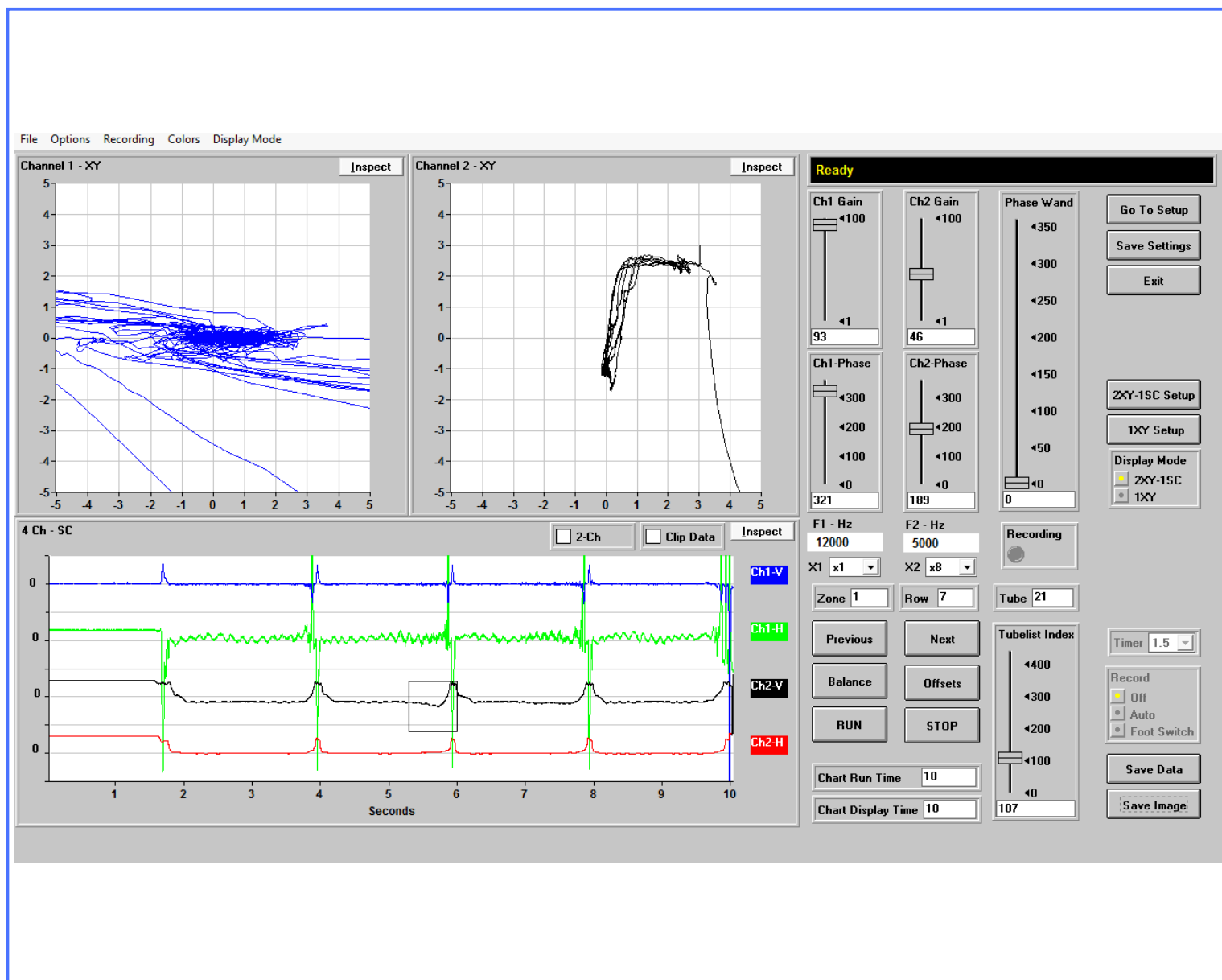
Right Hand Facing Controls



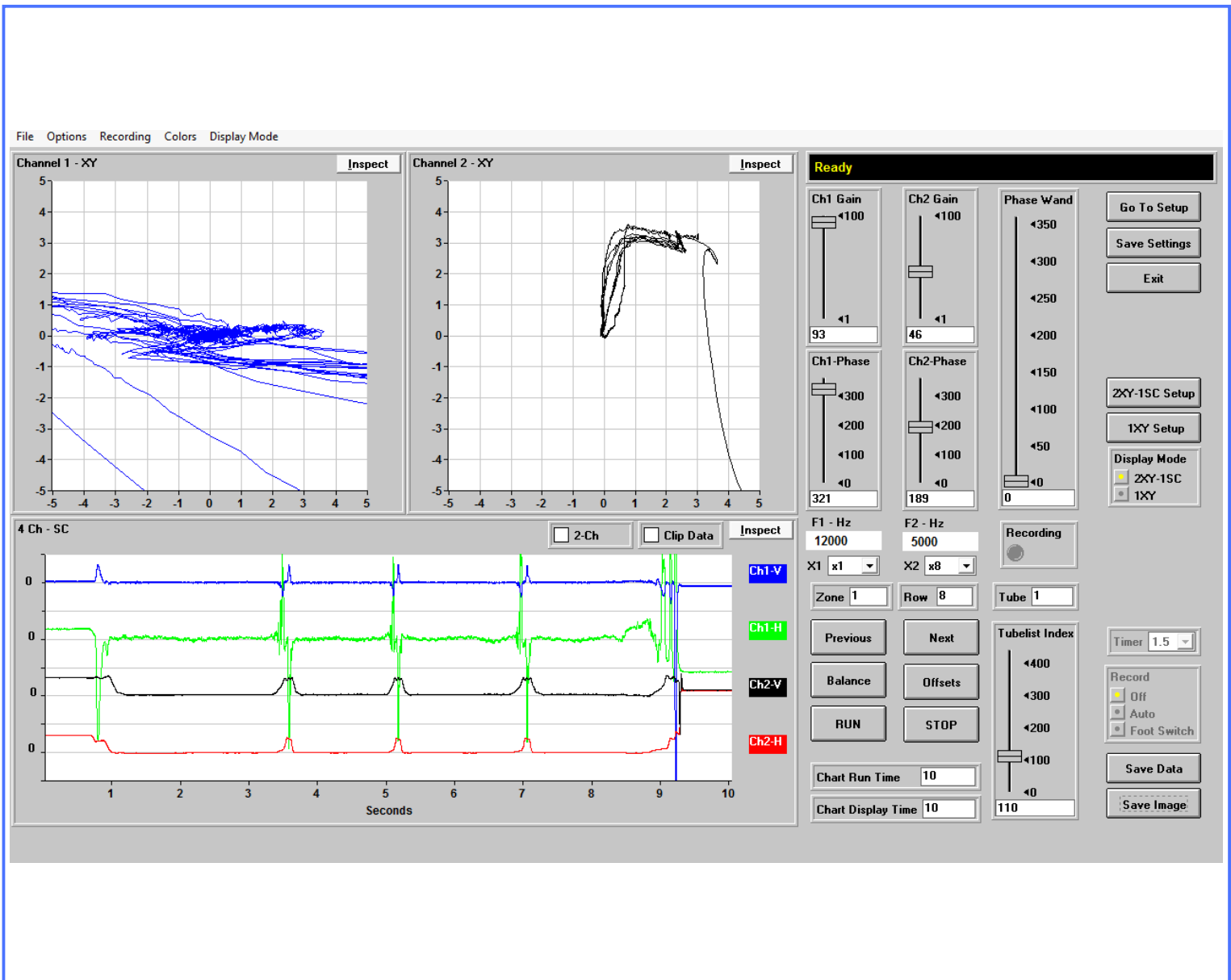
No Significant defects were found.

Condenser Section

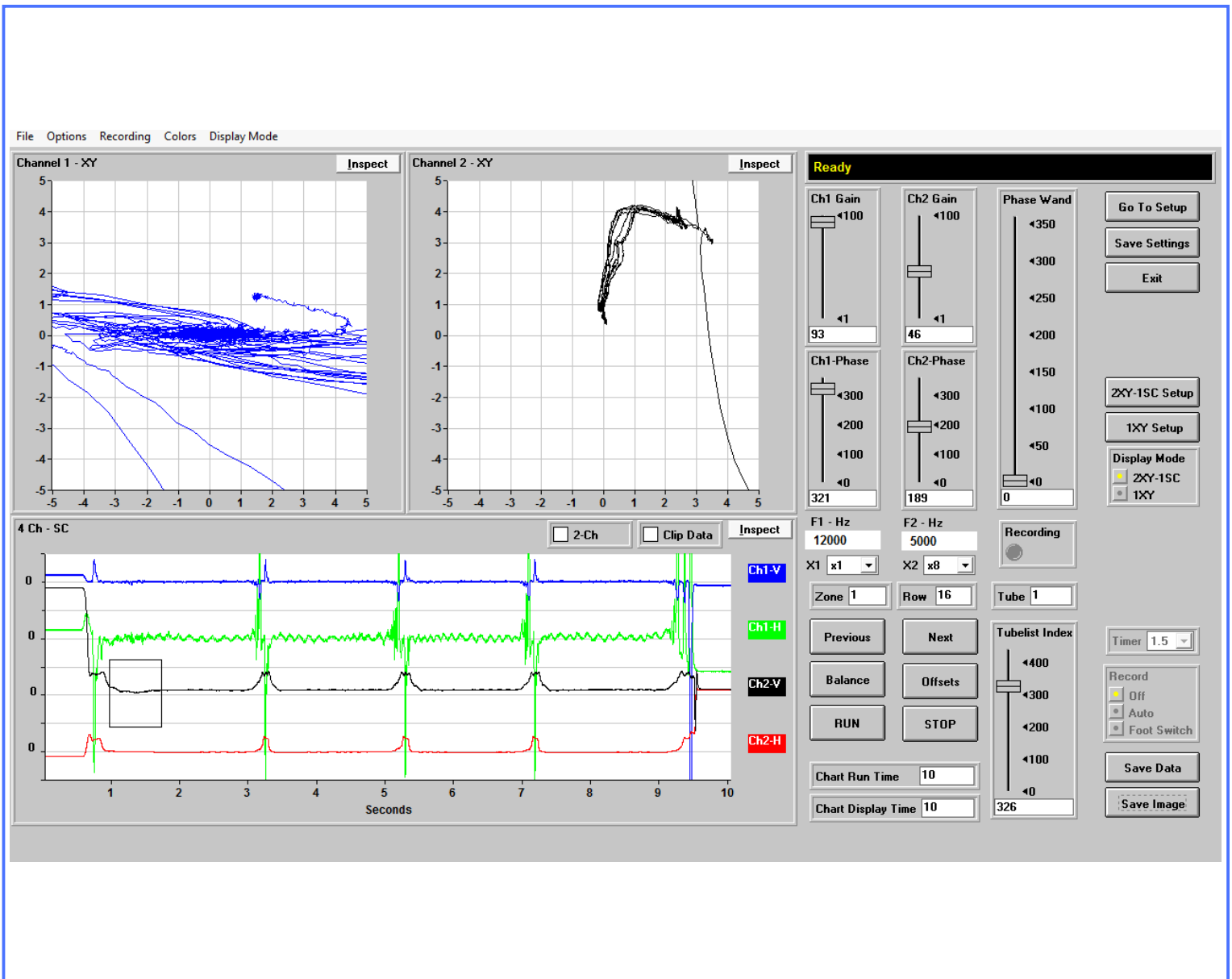
NO SIGNIFICANT DEFECTS (Row 1 Tube 1)

Condenser Section

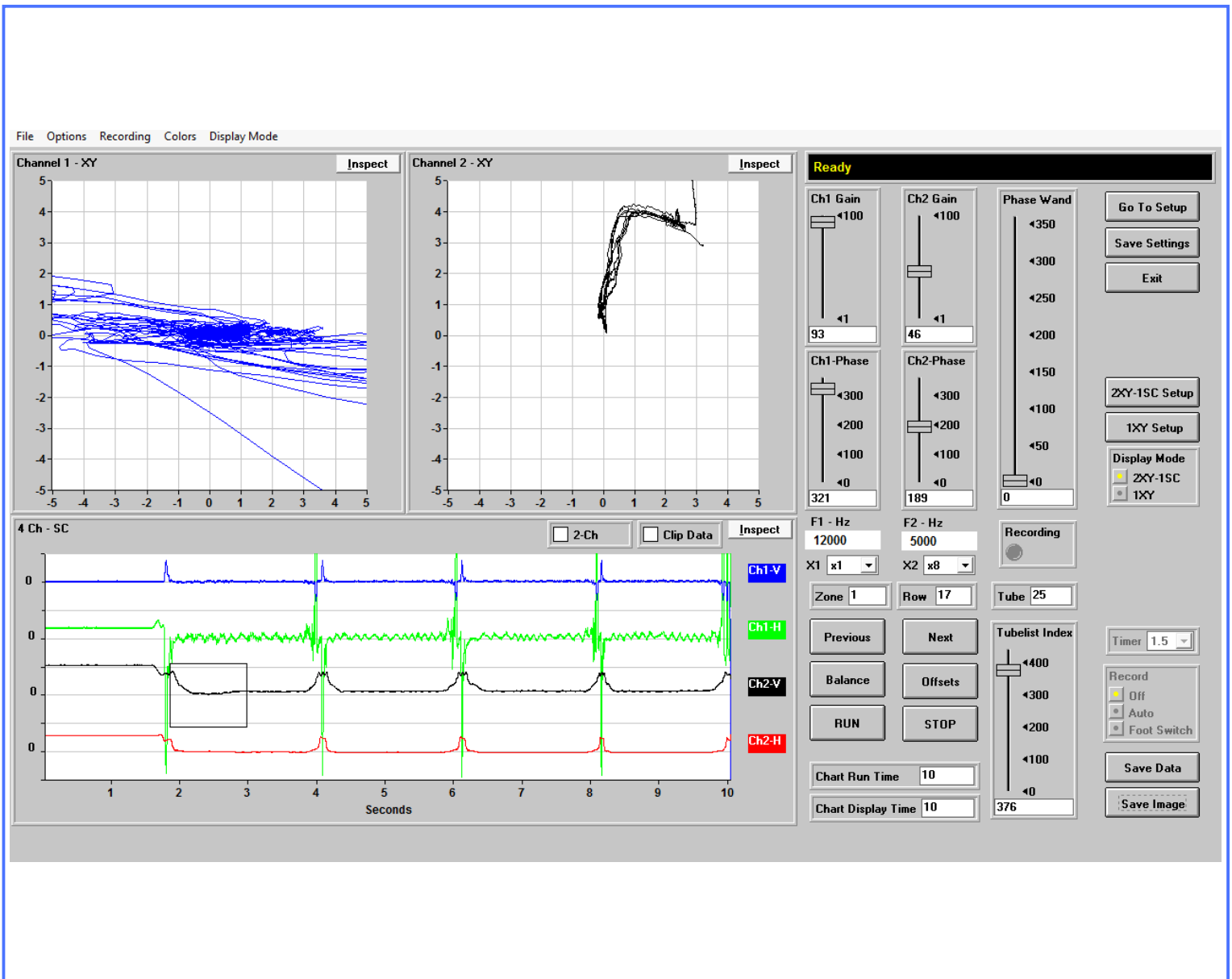
PLF 20% - 39% (Row 7 Tube 21)

Condenser Section

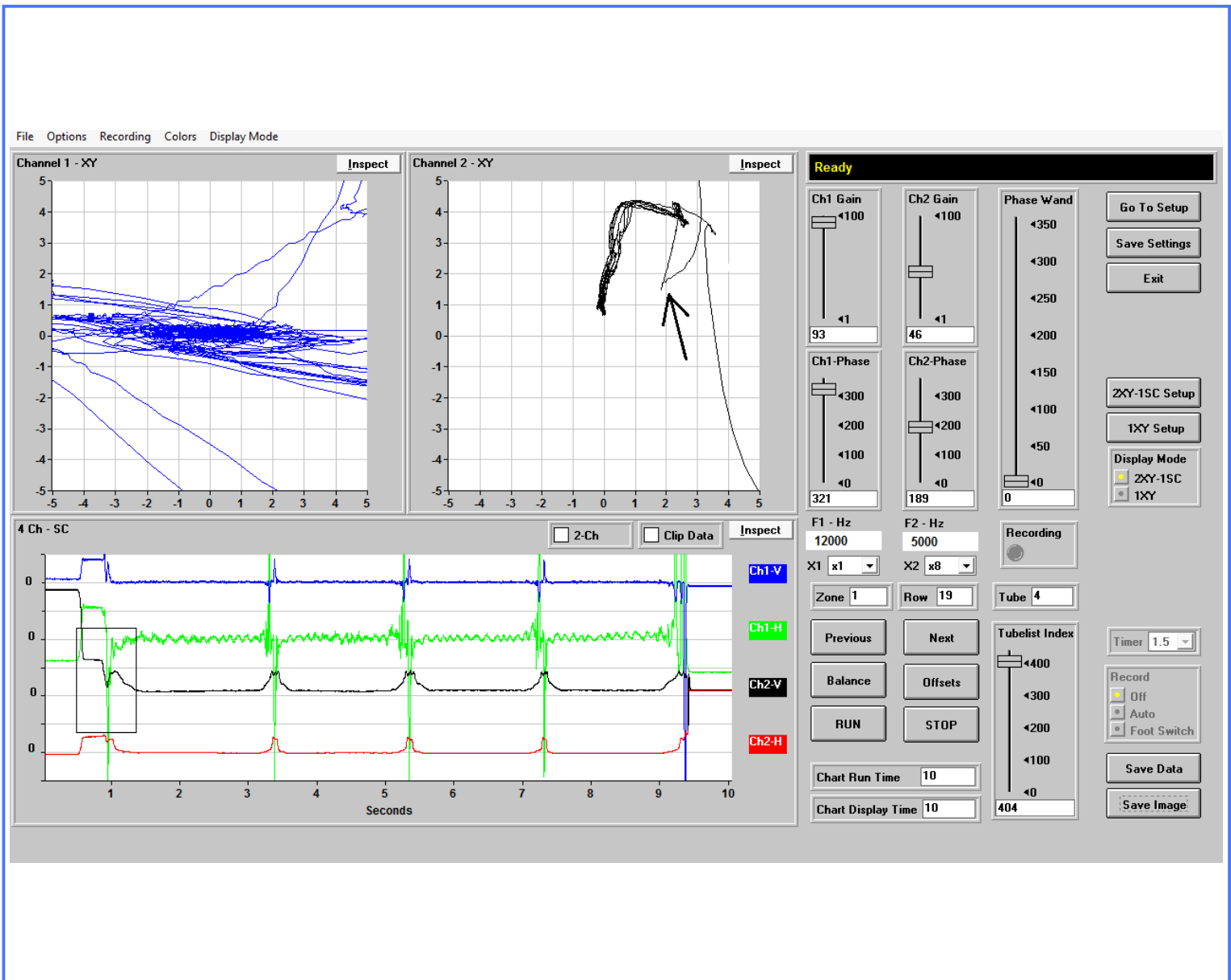
NO SIGNIFICANT DEFECTS (Row 8 Tube 1)

Condenser Section

PLF 20% - 39% (Row 16 Tube 1)

Condenser Section

PLF 20% - 39% (Row 17 Tube 25)

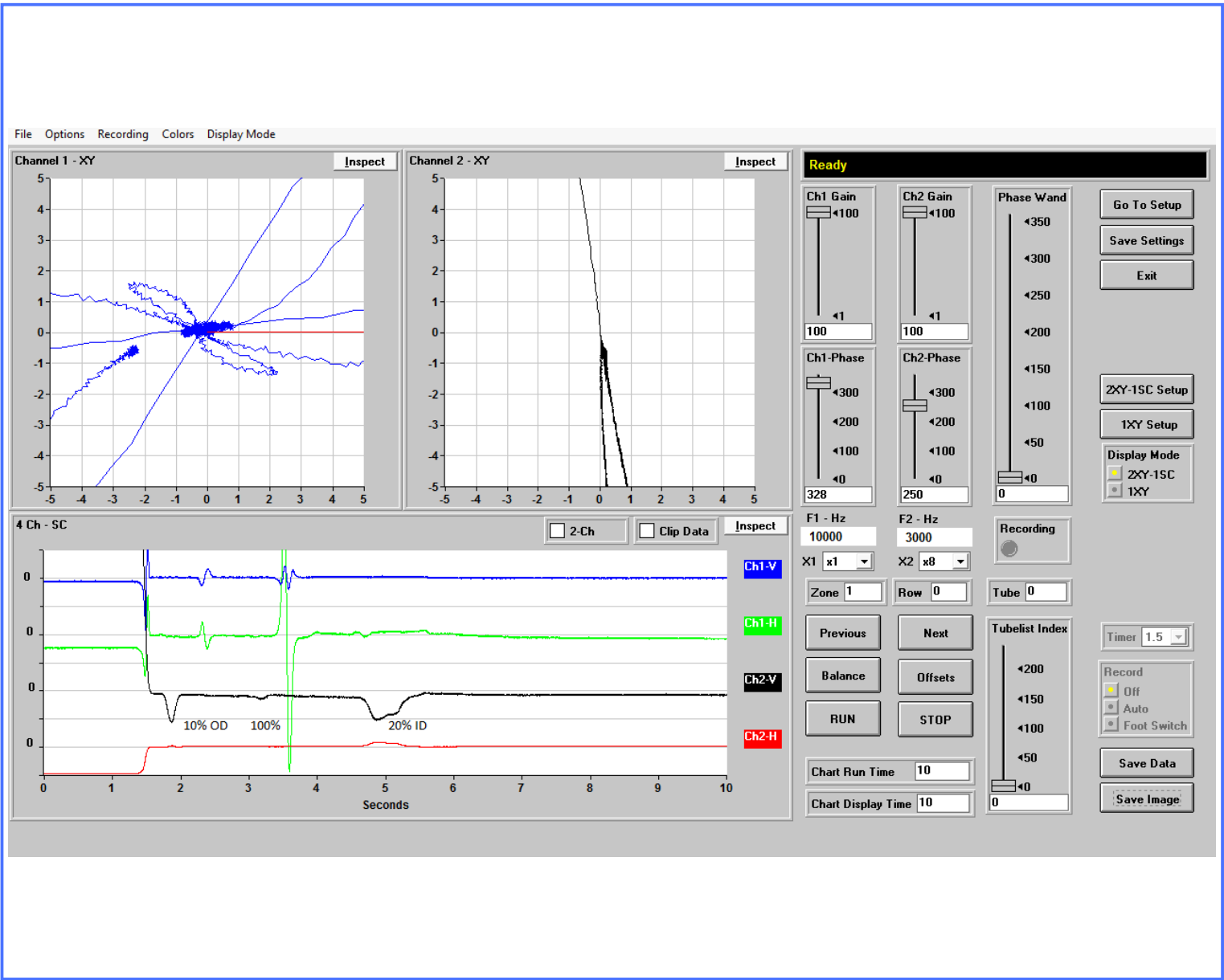
Condenser Section

ABNORMAL, NOMINAL (Row 19 Tube 4)

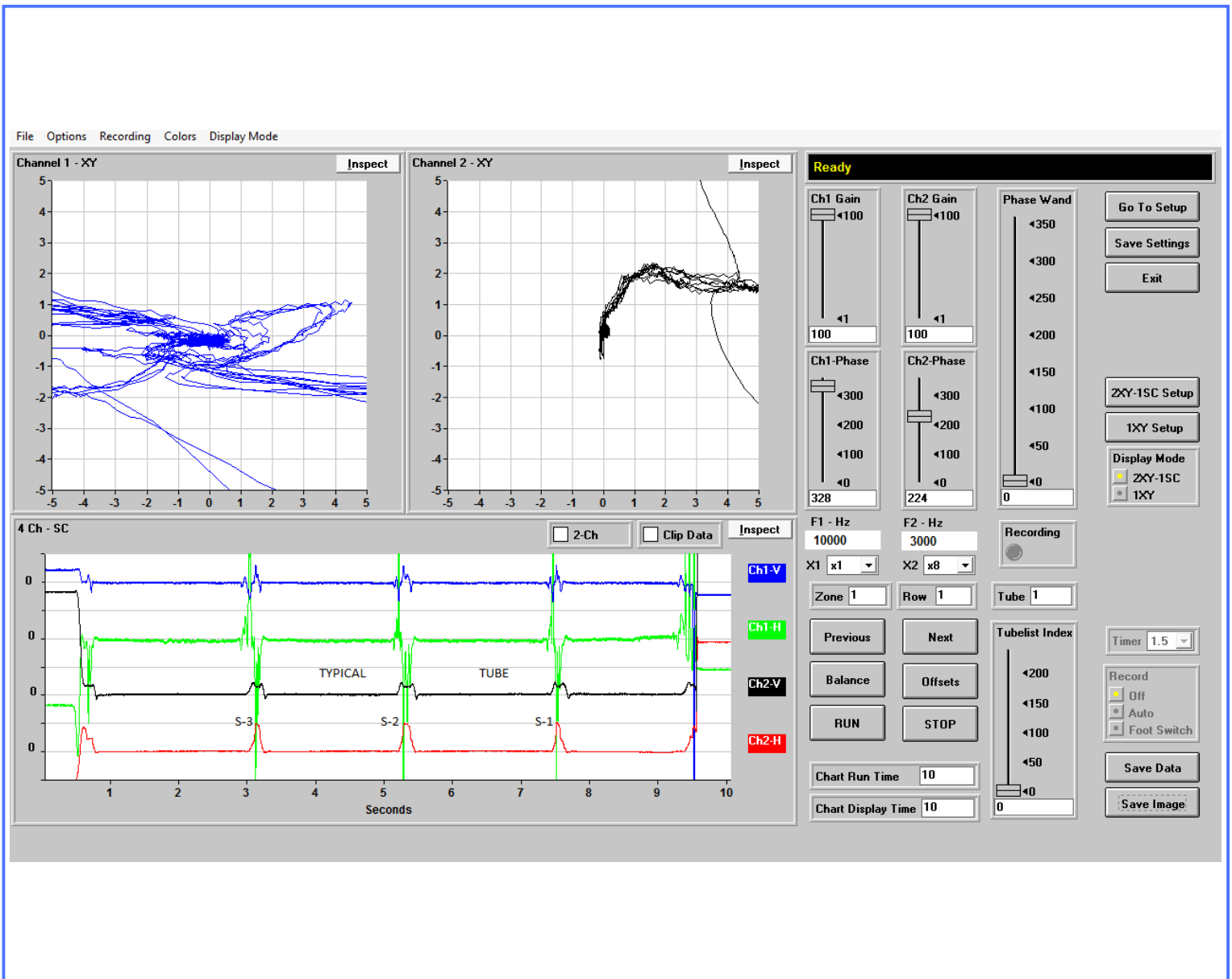
Calibration Page

Tube Type	Material	Nom Wall Thick	End Wall Thick	OD	Test Type	Probe Diameter
Skip Fin IE	Copper	.028	.053	1.000	CROSS/DIFF	.750

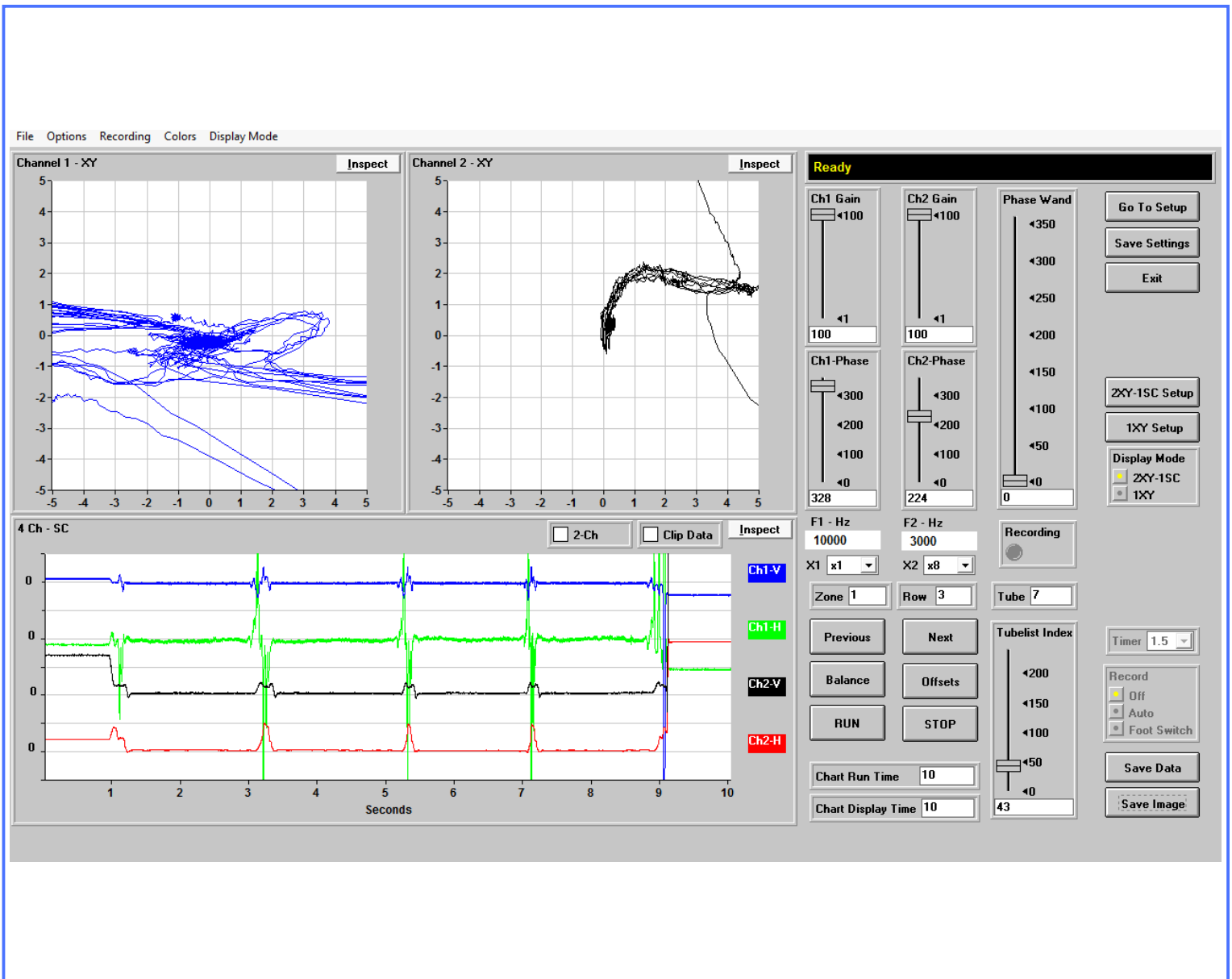
Condenser Evaporator



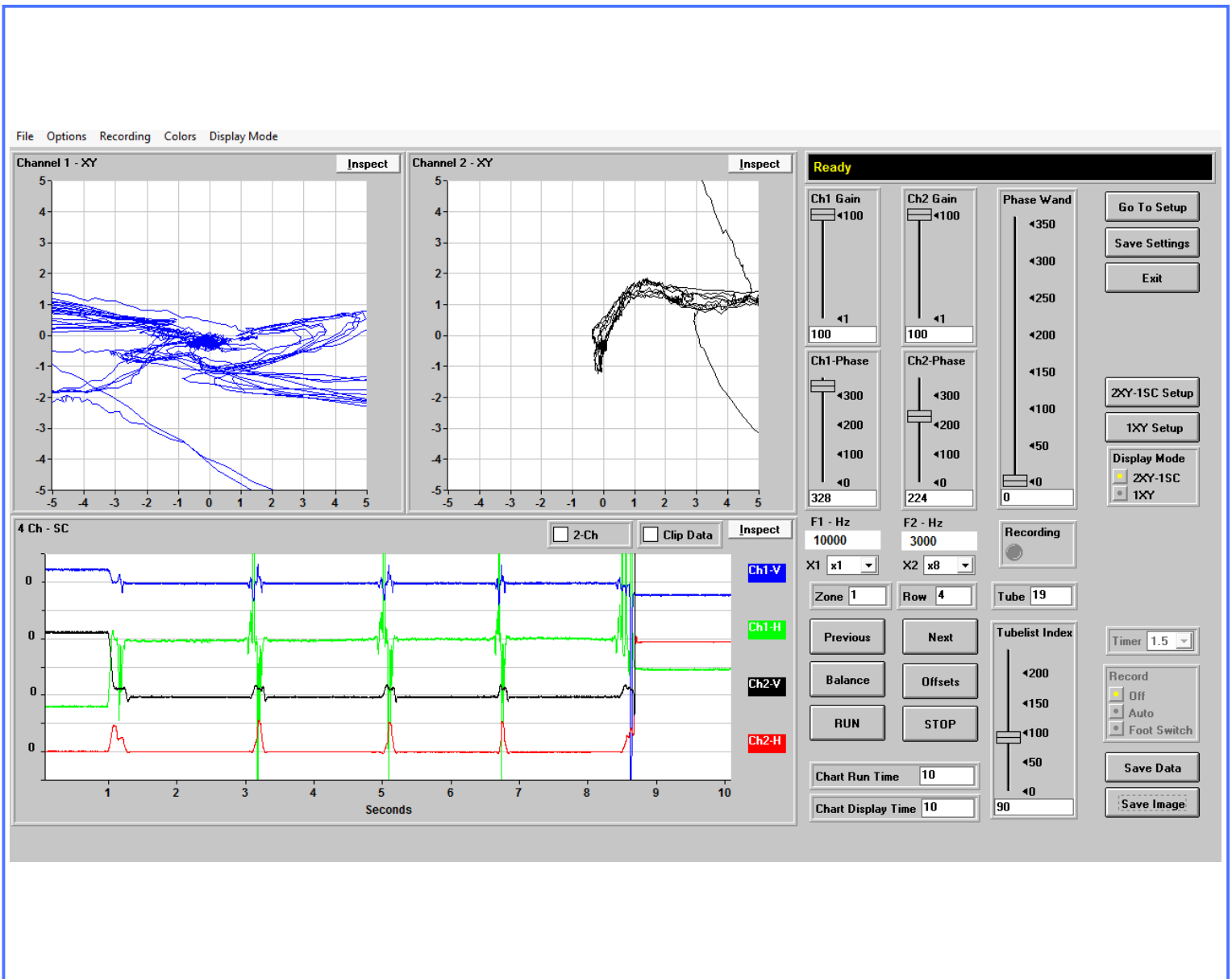
Note: Defects are compared to machined standards.
Actual Defect Geometry may differ.

Evaporator Section

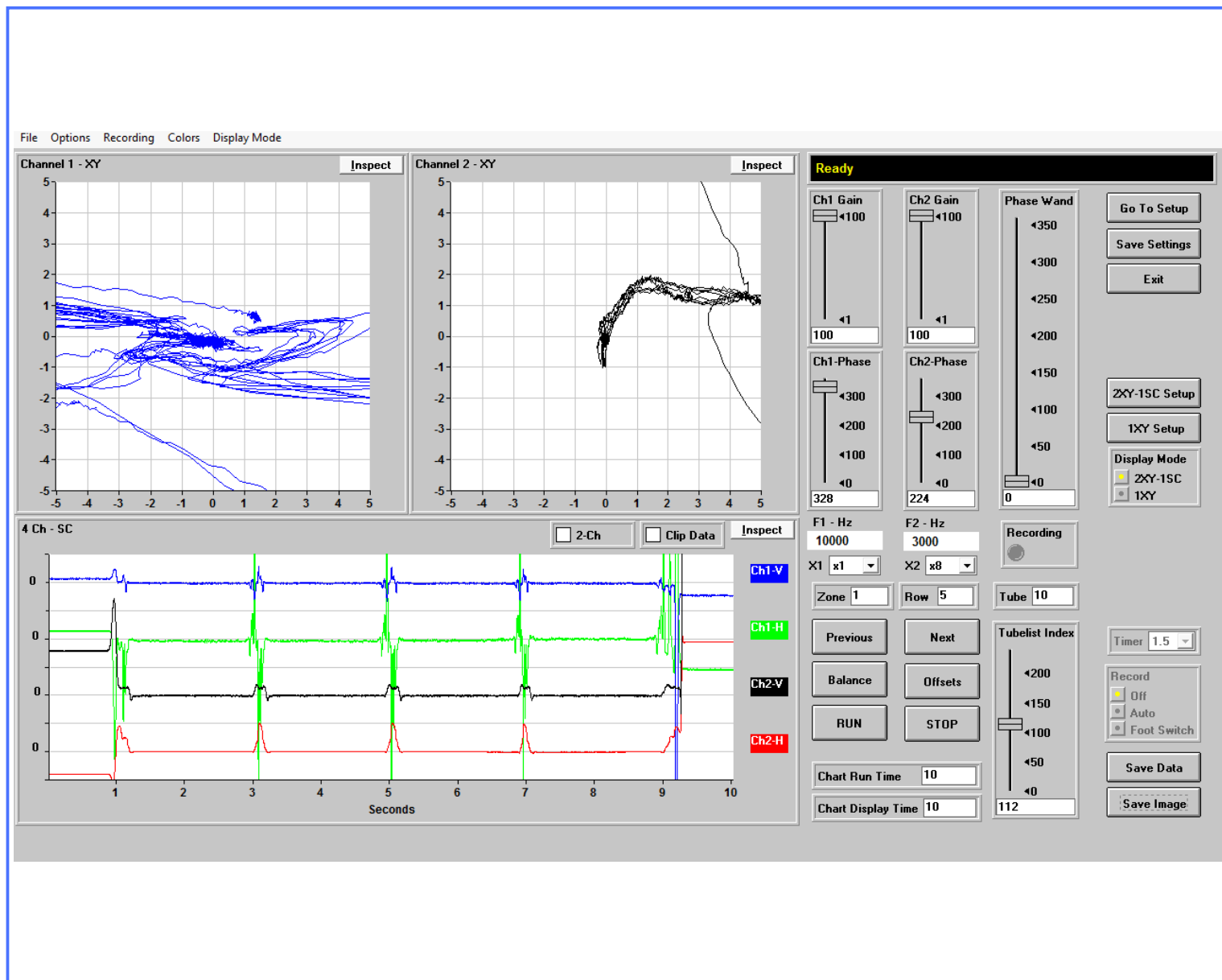
NO SIGNIFICANT DEFECTS (Row 1 Tube 1)

Evaporator Section

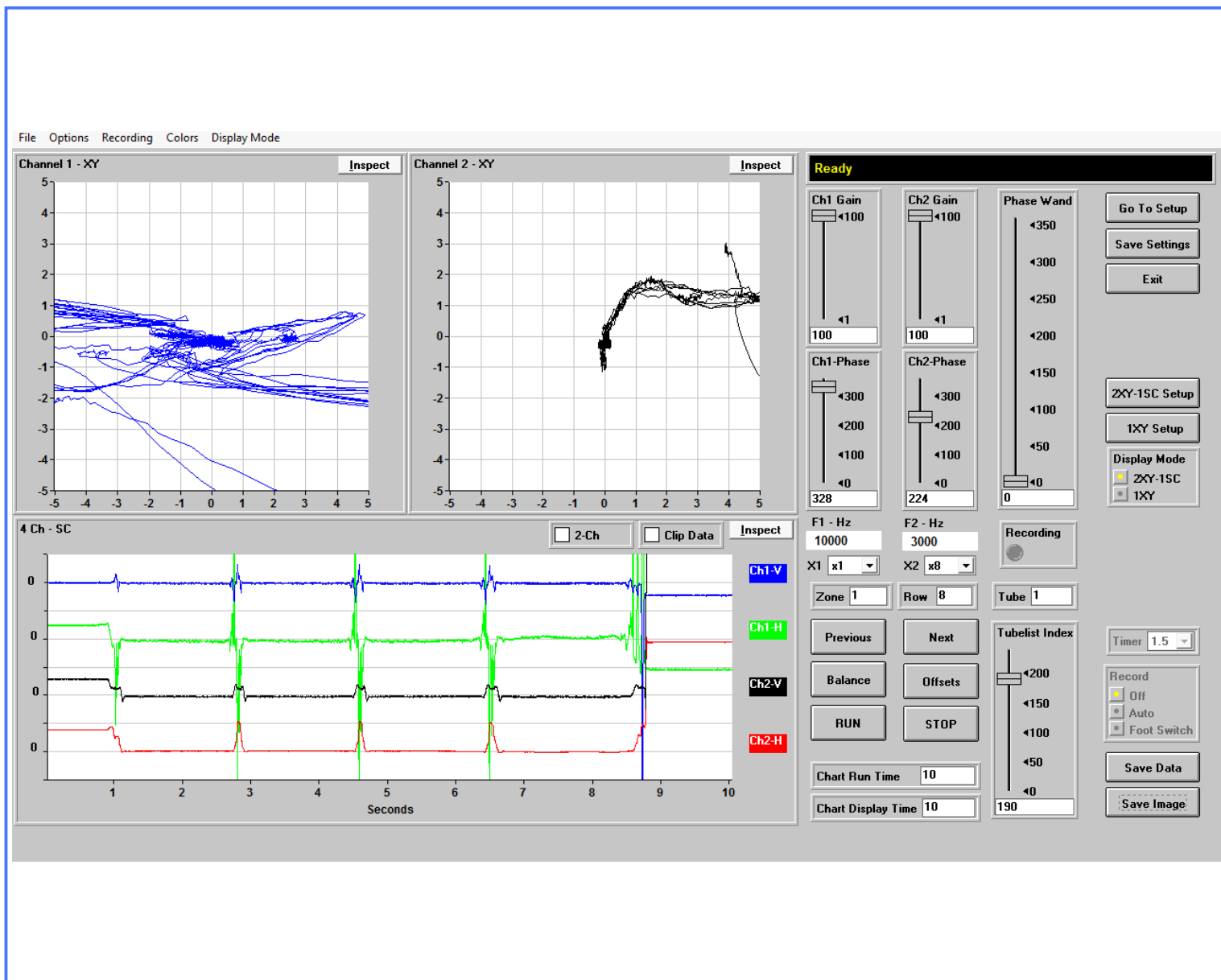
NO SIGNIFICANT DEFECTS (Row 3 Tube 7)

Evaporator Section

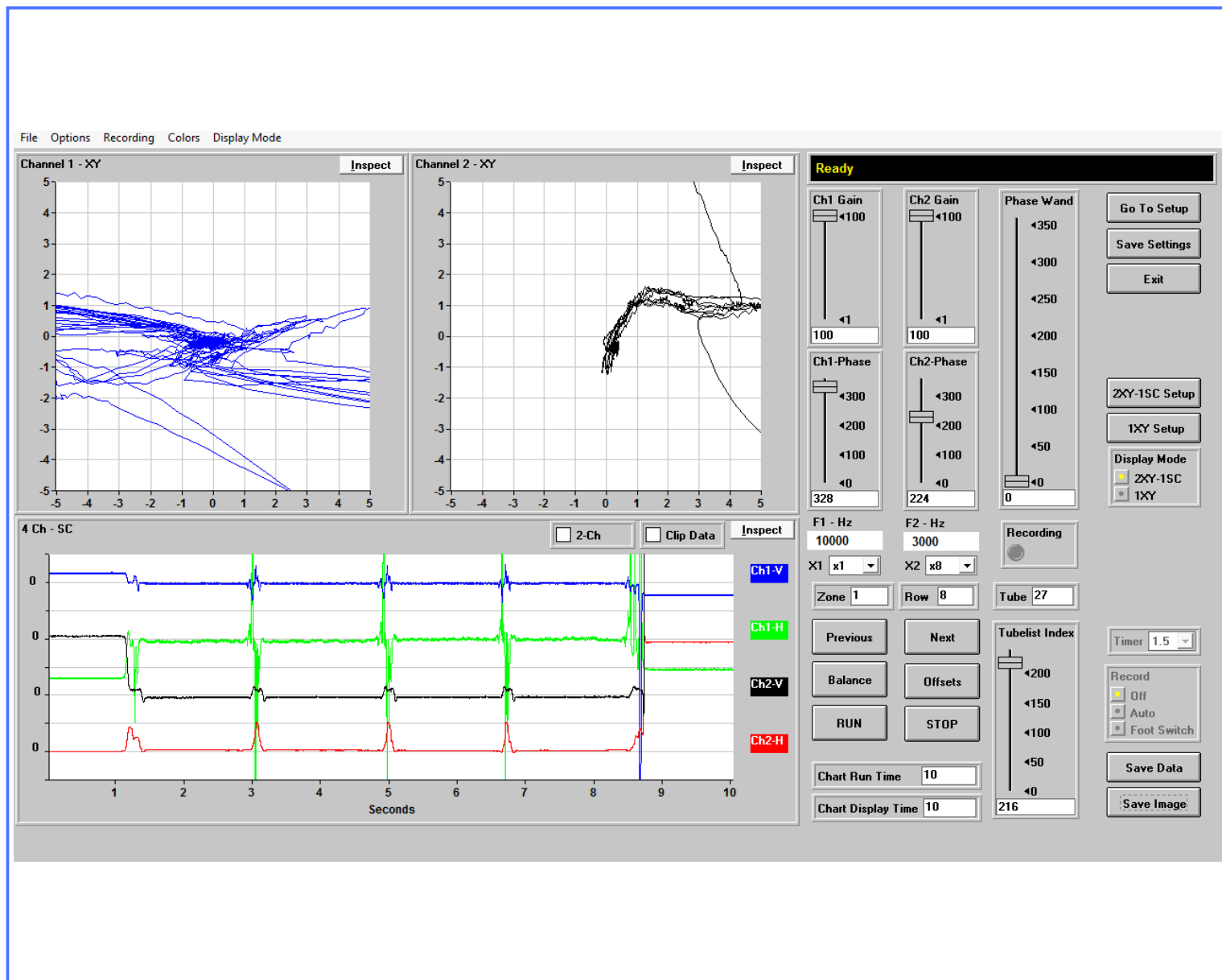
NO SIGNIFICANT DEFECTS (Row 4 Tube 19)

Evaporator Section

NO SIGNIFICANT DEFECTS (Row 5 Tube 10)

Evaporator Section

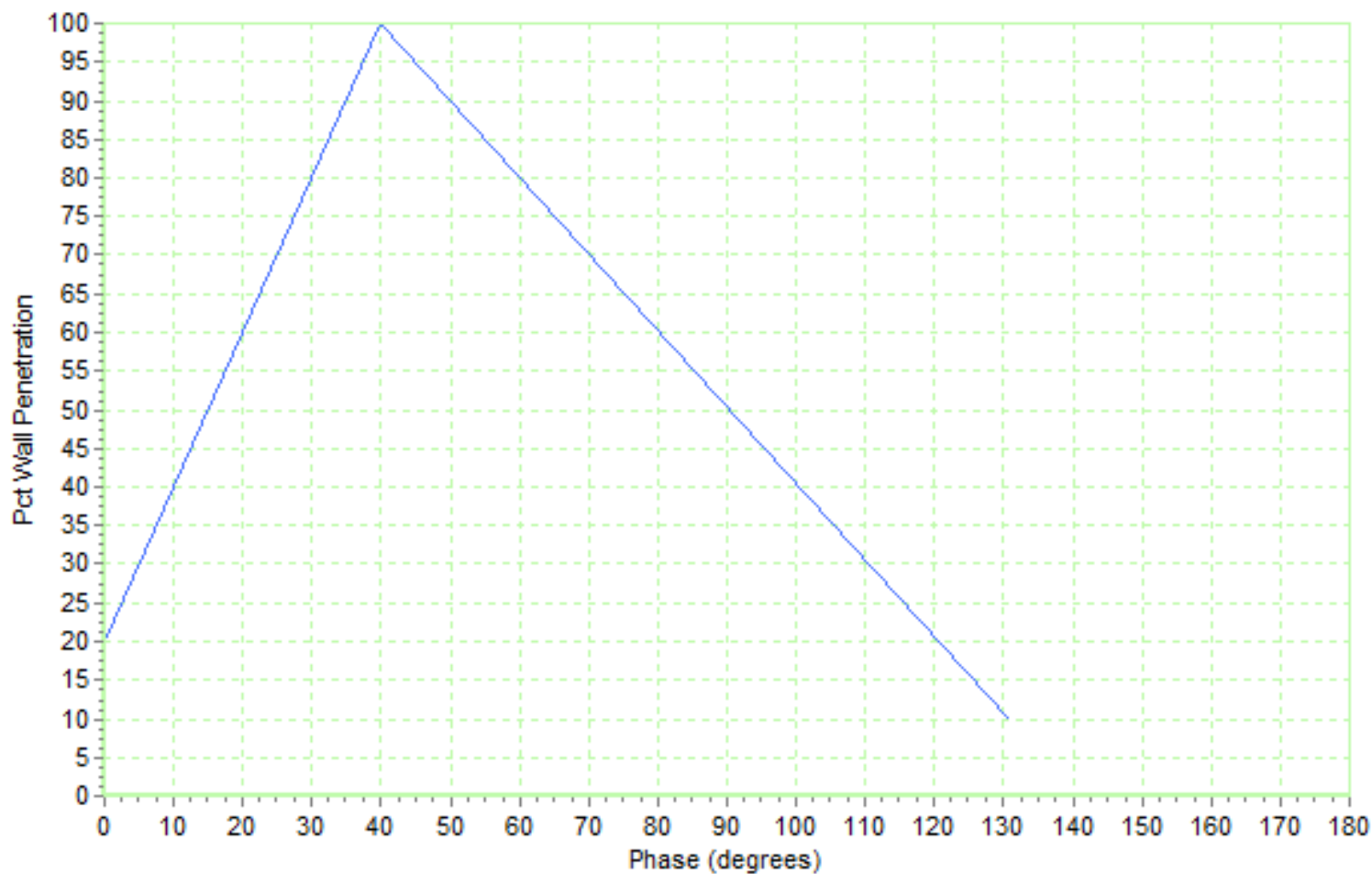
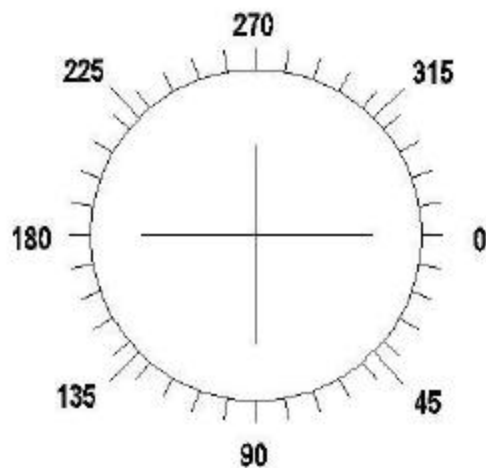
NO SIGNIFICANT DEFECTS (Row 8 Tube 1)

Evaporator Section

NO SIGNIFICANT DEFECTS (Row 8 Tube 27)

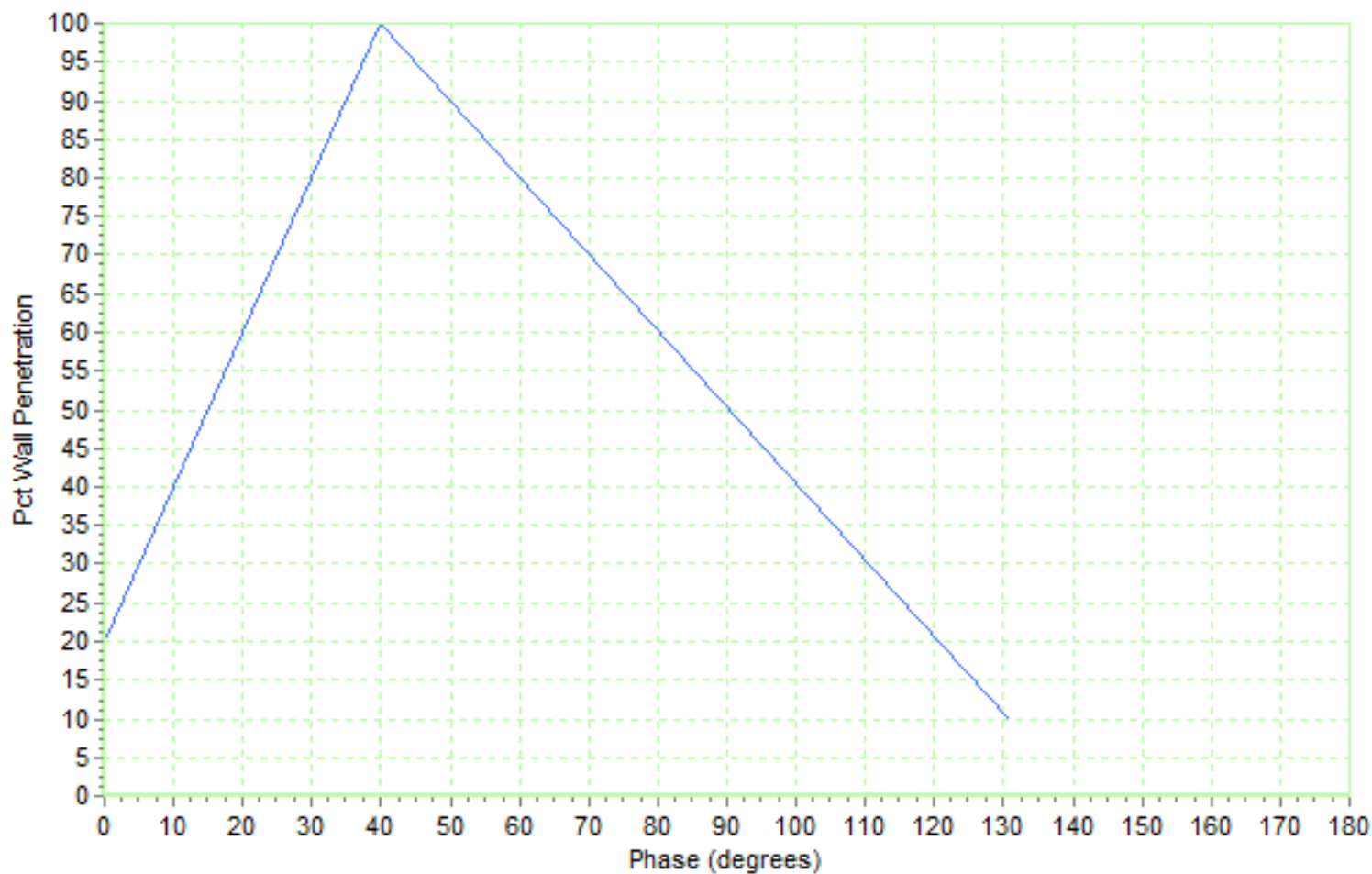
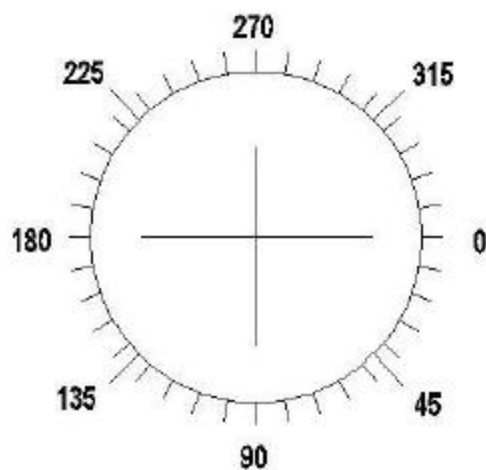
Phase Chart - Condenser

Material	Tube Type	OD	Wall	Test Type	Frequency	Probe Diameter
Copper	Skip Fin IE	1.000	Coppe	CROSS/DIFF	10,000	.028



Phase Chart - Evaporator

Material	Tube Type	OD	Wall	Test Type	Frequency	Probe Diameter
Copper	Skip Fin IE	1.000	.028	CROSS/DIFF	10,000	.750



Explanation of Abbreviations

Abbreviation	Explanation
ABN IND	Abnormal Indication
B	Bay
FB	Freeze Bulge
FBH	Flat Bottom Hole
FM	Foreign Material
ID	Internal Diameter
ID CORROSION	Internal Diameter, Corrosion
ID DEPOSIT	Internal Diameter, Deposit
ID PIT	Internal Diameter, Pit
IDML	Internal Diameter, Metal Loss
IE	Internally Enhanced
OD	Outside Diameter
ODML	Outside Diameter, Metal Loss
ODML@S	Outside Diameter Metal Loss at Support
OD DEPOSIT	Outside Diameter, Deposit
PLF	Possible Longitudinal Flaw
PRF	Possible Radial Flaw
PSC	Possible Stress Corrosion
S	Support
WAS	Wear at Support
>	Greater Than
<	Less Than
OTE	Opposite Test End
TE	Test End

Calibration Procedure

A calibration procedure is performed prior to an inspection, and is repeated every 2 hours, or whenever improper operation of the test instrument is suspected. Test frequencies are selected prior to an inspection through experimentation to achieve optimum phase separation, and amplitude response for the tube type and alloy being inspected. An appropriate inspection probe is selected based on tube type, wall thickness, and alloy. The inspection probe will have a minimum fill factor of 80% through the smallest areas of the tubes being inspected. Instrument sensitivity is set high enough to determine background noise inherent in the tube and to produce a .05 Volt deflection for a .031 through wall hole at .25 V/Div.

Calibration Reference Standard

A Calibration Reference Standard representing a typical production run tube of the same alloy, tube type and nominal wall thickness is used to adjust test system response. The calibration reference standard used for the inspection of finned and internally enhanced tubing, has been milled in accordance with the American Society for Testing and Materials (ASTM). Standard Recommended Practices, E-243-80, E-426-76, and E571-76. The depth of the grooves and notches used for establishing instrument response are calculated to compensate for the influence of the fins and/or internal enhancements used on finned tubes. Where applicable, calibration reference standards are milled in accordance with the American Society of Mechanical Engineers (ASME), Section V, Article 8, Appendix I.

A strip chart recording of each calibration reference standard used for the inspection has been included in this report. Each artificial discontinuity has been identified on the strip chart recording.